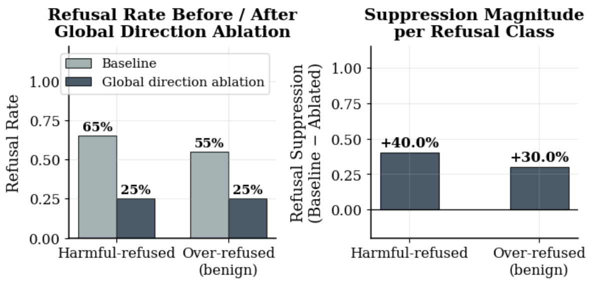

Figure 10: Effect of global ablation on harmful-refusal vs. over-refusal prompts. Ablation suppresses harmful-refusals more than over-refusal.

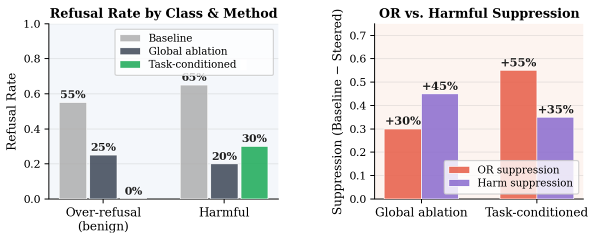

Figure 11: Global ablation vs. task-conditioned steering. Task-conditioned steering eliminates over-refusal while preserving more of the harmful-refusal mechanism.

subspace, suppresses over-refusal entirely while causing less disruption to the harmful-refusal mechanism than global ablation (Table 3; Figure 11). The suppression ratio from Table 3 suggests that the global ablation disrupts harmful-refusal more than it reduces over-refusal. This further suggests that task-dependent geometry is functionally consequential: it is measurable, and can be exploited. Global ablation fails for over-refusal mitigation as it operates on the wrong geometry.

4.6 Localising the Refusal Circuit

To test if any individual head is causally responsible, we apply Causal Patching (Meng et al., 2022; Conmy et al., 2023): we take a pair of prompts (one that causes a refusal, and the other that does not) and substitute the output of a single attention head from the non-refusing prompt into the refusing one at the final token position across all layer and observe whether the model changes its decision. If patching a head consistently flips the output, the specific head is a causal bottleneck; if no head does so, the decision is distributed. We patch the top attribution heads under three conditions: globally (all tasks combined), and per task for sentiment analysis and translation.

Table 4 report causal patching flip rates, the fraction of test pairs where replacing a head's output with a non-refusing counterpart flips the final refusal decision. The highest rates of 40%, appear

	Global ablation	Task-conditioned
OR: before → after	55%→25%	55%→0%
RH: before → after	65%→20%	65%→30%
Suppression (OR/RH)	0.67	1.57

Table 3: Comparison of Global Ablation and Task-conditioned methods on Over Refusal (OR) and Harmful-refusal (RH). Suppression = OR_reduction (%) / RH_reduction (%). Task-conditioned method from Maskey et al. (2025).

for a cluster of mid-layer heads (Layers 13 and 15) under the global condition. We find that no head crosses a sufficient causal-necessity threshold (e.g. 50%) under any condition.

Head	Condition	Flip rate
L13.H02	Global	2/5 (40%)
L13.H08	Global	2/5 (40%)
L13.H12	Global	2/5 (40%)
L13.H15	Global	2/5 (40%)
L13.H24	Global	2/5 (40%)
L15.H08	Global	2/5 (40%)
L22.H19	Global	1/5 (20%)
L30.H03	Translate	2/5 (40%)
L31.H13	Translate	1/5 (20%)
L31.H21	Translate	1/5 (20%)
(all heads)	Sentiment	0/4 (0%)

Table 4: Causal patching flip rates (Refusal to Non refusal) by head and condition. No head reaches a sufficient necessity threshold.

Figure 12 visualises these rates per task: the sentiment task produces no flips across all heads, while the translate condition shows modest, inconsistent flips for L30.H03 (40%) (Figure 12). Mechanistically, the most prominent object was the attention head on layer 30 (L30.H3) and geometrically steering this layer alone suppresses over-refusal from 55%→17.55%, which falls shorter than steering multiple mid-layers discussed earlier in §4.5. The result is not symmetric across tasks: if refusal were implemented through a shared bottleneck head, we would expect at least one head to flip consistently regardless of task.

The null result further supports the geometric finding that refusal is committed through distributed, multi-layer, task-specific circuits, where removing individual heads does not meaningfully alter the underlying signals, and that intervention

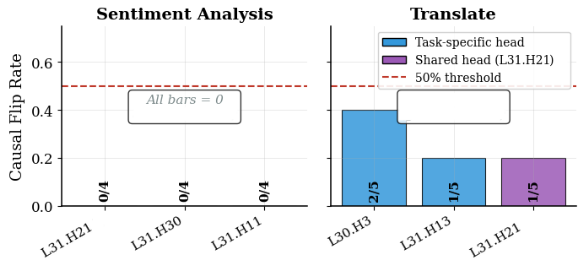

Figure 12: Per-task causal patching flip rates. Blue bars: task-specific attribution heads; purple bar: shared head L31.H21. Sentiment yields no flip; translate shows modest flips for a small subset of heads.

at the residual stream is necessary to achieve meaningful suppression. In line with feature analysis Lindsey et al. (2025), we suspect that layer 30 in LLaMA may conceal refusal-specific SAE features, with task-specific features developed in the mid-layers. This suggests that intervention at the residual-stream or feature-level is necessary to achieve meaningful suppression.

5 Discussion

Three-phase geometry of refusal. We discuss the layer-wise phases (early, mid and late layers) across task and refusal types: (a) geometrically on residual stream and (b) mechanistically by probing and patching. In early layers, our mechanistic analysis suggests that although the task identity is linearly decodable (§4.3), geometrically the residuals are not yet well formed (§4.2). In the mid-layers, task-identity clusters dominate and the representations organise by task, with over-refusal differentiable by a small distance within-task perturbation.

In the late layers, task-specific attention heads commit the final refusal decision distributedly, with almost no prominent attention head bottleneck. Steering the most mechanistically prominent layer 30 performed marginally better steering than the universal direction, but fell short compared to the multi-layers steering on mid-late layers. Task-identity cluster structure of comparable strength also replicates on Qwen1.5-7B-Chat (Appendix A.5), suggesting the geometry generalises across models.

Why global ablation is structurally insufficient. The global harmful-refusal direction $\hat{\mathbf{r}}_\ell$ is task-agnostic by construction: it is extracted from text samples that are harmful, without taking intent of the prompt or the task into consideration. We identified that over-refusal representations reside in a specialised region, organised around task-identity axes at mid-layers. The partial geometric overlap between the two directions causes incidental over-refusal suppression when one is ablated, but because they are not the same direction, this suppression is imprecise. The geometry does not permit a task-agnostic direction to decompose the task-conditioned over-refusal signal cleanly.

Scope of the linear representation hypothesis. We distinguish two senses in which refusal may be considered linear (Park et al., 2023): causal linearity, whether ablating a single direction produces meaningful behavioural change, and representational linearity, whether population variance is concentrated along one principal direction. For harmful-refusal, causal linearity holds but representational linearity does not: PC1 accounts for a third of the variance, so the intervention succeeds not because the geometry is simple, but because one axis happens to be causally sufficient. For over-refusal, neither condition is satisfied: the suppression ratio below one (Table 3) confirms causal linearity fails, and the subspace dimensionality analysis confirms representational linearity fails too, suggesting that the failure of global ablation follows directly from the representational geometry.

6 Conclusion

Harmful-refusal and over-refusal are two distinct phenomena with distinct geometry. Harmful-refusal directions are task-agnostic, and they converge in the early-to-mid layers of the network, and can be sufficiently addressed by a single global direction. Over-refusal directions are task-dependent: they live inside task-identity clusters, vary across tasks, and span a higher-dimensional subspace. Ablating the global refusal direction to fix over-refusal applies the wrong geometry: partial directional overlap produces incidental suppression, but it disrupts the safety mechanism more than addressing over-refusals. A task-conditioned method achieves substantially better precision, suggesting that the geometric differences between harmful-refusals and over-refusals are consequential.

The core finding is that global ablation, effective for harmful-refusal, fails on over-refusal because the two phenomena occupy geometrically distinct subspaces. Addressing over-refusal therefore requires methods that are sensitive to task-dependent structure, whether during training or fine-tuning, or through targeted mitigations in the residual stream.

Limitations

Our findings are limited to two instruction-tuned models (LLaMA and Qwen), and whether the geometric properties identified here generalise to larger models or different alignment procedures can be further explored. Per-task directional analysis is limited to the tasks where over-refusal occurs; the remaining tasks produce limited to no over-refusal samples, which restricts the scope of task-conditioned geometric analysis. The results from causal patching should be treated as indicative of distributed causal structure rather than precise effect estimates, given the small number of contrastive pairs per condition. The experiments analyse the geometry of residual streams, and could also be analysed at feature level using SAEs. Cross-model replication (Appendix A.5) confirms task-identity cluster structure on Qwen1.5-7B, but limitedly analyses the directional claims due to minimal refusal rates.

Ethical Considerations

This work is a mechanistic analysis of representational geometry and does not introduce new methods for bypassing safety mechanisms. The task-conditioned mitigation approaches operates only within pre-specified benign task subspaces and should not be applied outside developer-defined, high-confidence benign contexts. Use of such techniques should be coordinated by red-teaming to ensure that harmful-refusal rates are not adversely affected.

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